

Binomial Distribution

Question 1. Suppose that the probability that an adult person is suffering from diabetes in a population is 0.10. Suppose that we randomly select a sample of 6 persons.

- Find the probability distribution of the random variable (X) representing the number of persons with diabetes in the sample.
- Find the expected number of persons with diabetes in the sample (mean of X). [Ans: 0.6]
- Find the variance of X . [Ans: 0.54]
- What is the probability that there will be exactly 2 persons with diabetes? [Ans: 0.09842]
- What is the probability that there will be at most 2 persons with diabetes? [Ans: 0.98415]
- What is the probability that there will be at least 4 persons with diabetes? [Ans: 0.00128]

Solution: We are interested in the following random variable:

X = the number of persons with diabetes in the sample of 6 persons.

Notes:

- Bernoulli trial: diagnosing whether a person has diabetes or not. There are two outcomes for each trial
- Number of trials = 6, or $n=6$.
- Probability of success: = 0.10.
- Probability of failure:
- The trials are independent because of the fact that the result of each person does not affect the result of any other person since the selection was made at random.

The random variable, X , has a binomial distribution with parameters: $n=6$ and $p=0.10$, that is: $X \sim \text{Binomial}(n, p)$,

$X \sim \text{Binomial}(6, 0.10)$.

The possible values of X are $X=x$, where, $x = 0, 1, 3, 4, 5, 6$

[Try yourself]

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Question 2. Heart failure is due to either natural occurrences (87%) or outside factors (13%). Outside factors are related to induced substances or foreign objects. Natural occurrences are caused by arterial blockage, disease, and infection. Suppose that 20 patients will visit an emergency room with heart failure. Assume that causes of heart failure between individuals are independent.

- What is the probability that three individuals have conditions caused by outside factors?
- What is the probability that three or more individuals have conditions caused by outside factors?
- What are the mean and standard deviation of the number of individuals with conditions caused by outside factors?

Solution: Given:

Probability of heart failure due to outside factors: $p = 13\%$ or 0.13

Number of patients: $n = 20$

Let X be a random variable that represents the number of patients with heart failure due to outside factors. Therefore, X follows a binomial distribution i.e., $X \sim \text{Binom}(n = 20, p = 0.13)$. and the probability mass function (pmf) of X is given by

$$P(X = x) = {}^{20}C_x \times (0.13)^x \times (1 - 0.13)^{20-x}; \quad x = 0, 1, 2, \dots, 20$$

- The probability that three individuals have conditions caused by outside factors:

$$P(X = 3) = {}^{20}C_3 \cdot (0.13)^3 \cdot (1 - 0.13)^{20-3} \approx 0.2347$$

Interpretation: There is approximately a 23.47% chance that exactly 3 out of 20 patients will have heart failure caused by outside factors.

- The probability that three or more individuals have conditions caused by outside factors:

This can be calculated as:

$$P(X \geq 3) = 1 - P(X < 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

Calculate each term:

$$P(X = 0) = {}^{20}C_0 \cdot (0.13)^0 \cdot (1 - 0.13)^{20-0} \approx 0.0617$$

$$P(X = 1) = {}^{20}C_1 \cdot (0.13)^1 \cdot (1 - 0.13)^{20-1} \approx 0.1844$$

$$P(X = 2) = {}^{20}C_2 \cdot (0.13)^2 \cdot (1 - 0.13)^{20-2} \approx 0.2618$$

Now add them up:

$$P(X \geq 3) = 1 - (0.0617 + 0.1844 + 0.2618) = 1 - 0.5079 \approx 0.4921$$

Interpretation: There is approximately a 49.21% chance that three or more individuals out of 20 will have heart failure caused by outside factors.

- The mean and standard deviation of the number of individuals with conditions caused by outside factors: For a binomial distribution:

$$\text{Mean}, \mu = np = 20 \times 0.13 = 2.6$$

Standard deviation, $\sigma = \sqrt{np(1-p)} = \sqrt{20 \times 0.13 \times 0.87} = \sqrt{2.262} \approx 1.504$

Interpretation: On average, we expect about 2.6 patients out of 20 to have heart failure due to outside factors, with a standard deviation of approximately 1.504 patients.

Question 3. Suppose you have a fair spinner divided into four equal sections numbered 1 through 4, and a fair coin with two sides: heads (H) and tails (T). You spin the spinner and flip the coin simultaneously, and you record the number on the spinner and the result of the coin flip. Let's define a random variable Y as the number of times the spinner lands on 3 and the coin lands on heads (H) simultaneously. You conduct this experiment 15 times independently.

- Write down the probability distribution function for this scenario.
- Use the probability distribution function from (a) to determine the probability of getting the outcome (spinner lands on 3 and coin lands on H) "at least four times". Interpret your findings.
- Determine the mean and standard deviation of this probability distribution function.

Solution:

- Let us calculate the probability of getting a 3 on the spinner and heads (H) on the coin in a single trial.

Probability of spinner landing on 3, $P(3) = 0.25$

Probability of coin landing on heads (H), $P(H) = 0.5$

Therefore, probability of occurring both events (Spinner = 3 and Coin = H) in a single trial, $P(Y) = P(3) \times P(H) = 0.125$. Since the experiment was conducted 15 times independently, it follows binomial distribution with parameter, $n = 15$, $p = 0.125$. Thus, the probability mass function can be written as,

$$P(Y = y) = {}^{15}C_y 0.125^y (1 - 0.125)^{n-y}$$

- $P(Y \geq 4) = 1 - P(Y < 4) = 1 - (P(0) + \dots + P(3)) = 0.10778$

Comment: This probability 0.10778 represents the likelihood of getting the outcome of both the spinner landing on 3 and the coin landing on heads at least four times in 15 trials.

- [Hints: Mean and Variance of binomial distribution]

Comment: On average, we can expect the spinner to land on 3 and the coin to land on heads about 1.875 times in 15 trials, with a standard deviation of approximately 1.28, which measures the variability of this outcome.

Question 4. A factory produces items and has a 12% defect rate. If an inspector randomly selects 15 items.

- Calculate the probability of exactly 2 defective items. [Ans: 0.2870]
- Calculate the mean number of defective items expected [Ans: 1.8]
- Calculate the variance of the number of defective items. [Ans: 1.584]
- Determine the probability of having at most 2 defective items. [Ans: 0.7346]

Question 5. At BRAC University, the probability that a randomly selected student is enrolled in the course STA201 is 0.4. Suppose 20 students are randomly chosen from the university.

- What is the probability that exactly 8 out of these 20 students are enrolled in STA201? [Ans: 0.17971]
- What is the probability that fewer than 2 out of these 20 students are enrolled in STA201? [Ans: 0.00519]
- Calculate the mean and standard deviation of the binomial distribution representing the number of students enrolled in STA201 out of the 20 selected students. [Ans: Mean = 8, SD = 2.19]

Question 6. Based on data collected by the Bracu authority, an estimate of the percentage of students who have at some point in their life been told they have hypertension is 23.53 percent. If we select a simple random sample of 20 students and assume that the probability that each has been told that he or she has hypertension is .24, find the probability that the number of students in the sample who have been told that they have hypertension will be:

- Exactly three. [Ans: 0.14838]
- Fewer than three. [Ans: 0.10855]
- Between three and seven, inclusive. [Ans: 0.808]
- How many students who have been told that they have hypertension would you expect to find in a sample of 20? [Ans: 4.8]

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Question 7. The proportion of people with blood groups O, A, B and AB in a particular population are in the ratio 48:35:12:5, respectively. A random sample of 20 people from the population is picked.

- What is the probability that exactly 10 with blood group O?
- What is the probability that at most 2 with blood group AB?
- Find the expected number of people with blood group A. [Ans: 7]

Solution:

Given that, $O:A:B:AB = 48:35:12:5$

$$P(O) = 48/(48 + 35 + 12 + 5) = 0.48$$

$$P(A) = 0.35$$

$$P(B) = 0.12$$

$$P(AB) = 0.05$$

- $P(O = 10) = \binom{20}{10} (0.48)^{10} (1 - 0.48)^{20-10} = 0.1734$
- $P(AB \leq 2) = \binom{20}{0} (0.05)^0 \cdot (1 - 0.05)^{20-0} + \binom{20}{1} (0.05)^1 \cdot (1 - 0.05)^{20-1} = 0.92452$
- Try yourself.

Question 8. In a particular section of the university, 70% of students are from the CSE department, others are from the BBA department. If 15 students are randomly selected then,

- Write down the probability distribution function for this scenario. [$P(Y = y) = {}^{15}C_y (0.7)^y (0.3)^{15-y}$]
- Calculate the mean number of students from CSE department. [Ans: 10.5]
- Calculate the variance of the number from CSE department. [Ans: 3.15]
- Determine the probability of having at exactly 12 Student from CSE department. [Ans: 0.17]

Question 9. An MCQ test has 10 questions each with 5 possible choices. A student did not study at all and chose all the answers randomly.

- Probability that 2 answers will be correct. [$n = 10, p = 1/5$, Ans: 0.30199]
- Probability that at least 1 answer will be correct. [Ans: 0.89263]
- Probability that at most 2 answers will be correct. [Ans: 0.6778]
- Probability that less than 1 answer will be correct. [Ans: 0.10737]
- Average number of correct answers. [Ans: 2]
- Variance of this distribution. [Ans: 1.6]

Question 10. 70% of the passengers who travel on the newly launched “Korean Coach” to Cox’s Bazar from Dhaka buy ‘Bangladesh Protidin’ at the bookstall before they board the train. The train is full and each compartment holds 8 passengers.

- What is the probability that all the passengers in a compartment have brought the newspaper? [Ans: 0.058]
- What is the probability that none of the passengers in a compartment has brought the newspaper? [Ans: 6.561×10^{-5}]
- What is the probability that at most two passengers have brought the newspaper? [Ans: 0.011]
- Calculate expected number of passengers who have brought the newspaper. [Ans: 5.6]

Question 11. In a rural district of Bangladesh, a Hepatitis-B vaccination campaign was conducted to improve public health. It is known from previous health records that the Hepatitis-B inoculation is effective in 80% of cases. After 6 months, a health survey randomly selects 5 individuals from the vaccinated population to assess the effectiveness of the vaccination.

- Determine the probability that exactly one out of the 5 selected individuals did not benefit from the Hepatitis-B inoculation.
- Determine the probability that at least one of the 5 selected individuals did not benefit from the Hepatitis-B inoculation.
- Find the expected number of individuals out of the 5 who did not benefit from the Hepatitis-B inoculation.
- Calculate the variance and standard deviation of the number of individuals out of the 5 who did not benefit from the Hepatitis-B inoculation.

Solution:

Let X be the random variable representing the number of individuals out of 5 for whom the inoculation is not effective. Here, X follows a binomial distribution with:

Number of trials $n=5$

Probability of success (inoculation being effective) $p = 0.80$

Probability of failure (inoculation not being effective) $q = 1 - p = 0.20$

- For $k = 1$, $n = 5$, and $p = 0.80$

$$P(X = 1) = \binom{n}{k} p^{n-k} q^k = 0.409$$

- $P(X \geq 1) = 1 - P(x = 0) = 0.67$

- $E(x) = n \cdot q = 1$

- $V(x) = n \cdot p \cdot q = 0.80$

Poisson Distribution

Question 1. In a city, the average number of traffic accidents at a specific intersection is 5 per day.

- Determine the probability of having exactly 2 accidents on a given day.
- Find the probability of having at least two accidents on a given day.
- Find the probability of having at most one accident on a given day.
- Find the probability of having no more than three accidents on a given day.

Solution: Given that the average number of traffic accidents at a specific intersection is 5 per day, we can model the situation using a Poisson distribution. Let X be the random variable representing the number of accidents on a given day. Since X follows a Poisson distribution with mean $\lambda = 5$, the probability mass function of X is given by:

$$P(X = x) = \frac{e^{-5} 5^x}{x!}, x = 0, 1, 2, \dots$$

- Probability of having exactly 2 accidents on a given day:

$$P(X = 2) = \frac{e^{-5} 5^2}{2!} \approx 0.0842$$

Interpretation: There is an 8.42% chance of exactly 2 accidents occurring on a given day.

- Probability of having at least two accidents on a given day:

The probability of at least two accidents is the complement of having fewer than two accidents:

$$P(X \geq 2) = 1 - P(X < 2)$$

$$P(X < 2) = P(X = 0) + P(X = 1)$$

$$P(X = 0) = \frac{e^{-5} 5^0}{0!} \approx 0.0067$$

$$P(X = 1) = \frac{e^{-5} 5^1}{1!} \approx 0.0337$$

$$P(X \geq 2) = 1 - (0.0067 + 0.0337) \approx 0.9596$$

Interpretation: There is a 95.96% chance of having at least two accidents on a given day, indicating that it's highly likely more than one accident will occur.

- Probability of having at most one accident on a given day:

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

$$P(X \leq 1) = 0.0067 + 0.0337 \approx 0.0404$$

Interpretation: There's only a 4.04% chance of having at most one accident, making it unlikely that a day will pass with one or fewer accidents.

- d. Probability of having no more than three accidents on a given day:

$$P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$

$$P(X = 3) = \frac{e^{-3} 3^3}{3!} \approx 0.1404$$

$$P(X \leq 3) = 0.0067 + 0.0337 + 0.0842 + 0.1404 \approx 0.2650$$

Interpretation: There is a 26.5% chance of having no more than three accidents on a given day, showing that it's somewhat likely but still less than half the time.

Question 2. A bookstore sells 27 novels in 9 hours. Assume the number of novel selves follows Poisson distribution.

- Find the probability of selling exactly 5 novels in an hour.
- What is the probability of selling 0 novels in an hour?
- Calculate the mean and variance of novels sold per hour.
- What is the probability of selling more than 4 novels in an hour?

Solution:

- a. Find the probability of selling exactly 5 novels: $\lambda = 3, k = 5$

$$P(X = 5) = \frac{3^5 e^{-3}}{5!} = \frac{243 e^{-3}}{120} \approx 0.10082$$

- b. Probability of selling 0 novels:

$$P(X = 0) = \frac{3^0 e^{-3}}{0!} = e^{-3} \approx 0.0498$$

- c. Mean and variance of novels sold: $\mu = \lambda = 3, \sigma^2 = \lambda = 3$

- d. Probability of selling more than 4 novels:

$$\begin{aligned} P(X > 4) &= 1 - P(X \leq 4) \\ &= 1 - (P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)) = 0.18474 \end{aligned}$$

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Question 3. A library maintains records of daily visitor counts. On average, 10 people visit the library every hour. However, due to varying schedules and interests, the number of visitors can fluctuate. The library is open for 8 hours each day, and a study has shown that the number of visitors per hour follows a discrete distribution with a mean of 10 visitors per hour.

- Write down the probability mass function of the distribution.
- What is the probability that exactly 8 people visit the library in a given hour?
- What is the probability that at least 5 people visit the library in a given hour?
- What is the probability that the library gets fewer than 4 visitors in a two-hour period?
- What is the average number of visitors per hour and the standard deviation of the number of visitors per hour?

Solution:

- Let, X = Number of visitors per hour, and it follows Poisson distribution with parameter $\lambda = 10$. Thus, the distribution function can be written as:

$$P(X) = \frac{e^{-10} 10^x}{x!}$$

- $P(X = 8) = 0.1126$

Comment: The probability that exactly 8 people visit the library in a given hour is approximately 0.1126.

- $P(X \geq 5) = 1 - P(X < 5) = 0.9707$

Comment: The probability that at least 5 people visit the library in a given hour is approximately 0.9622.

- Here, $\lambda = 2 \times 10$; [Two-hour period]

$$P(X < 4) = P(0) + \dots + P(3) = 0.0000032$$

Comment: The probability that the library gets fewer than 4 visitors in a two-hour period is approximately 0.0000032.

- Try yourself.

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Question 4. During a recent awareness campaign in Dhaka, the emergency response center received an average of 8 emergency calls per hour related to the event.

- What is the probability that the center will receive exactly 7 emergency calls in an hour? [Ans: 0.1396]
- What is the probability that the center will receive at most 5 emergency calls in an hour? [Ans: 0.1912]
- What is the probability that the center will receive more than 6 emergency calls in an hour? [Ans: 0.6866]
- What is the probability that the center will receive exactly 24 emergency calls in a span of 4 hours? [Ans: 0.03 (when $\lambda = 32$)]

Question 5. The number of serious gastrointestinal reactions reported to the British Committee on Safety of Medicine was 538 for 9,160,000 prescriptions of the anti-inflammatory drug piroxicam. This corresponds to a rate of .06 gastrointestinal reactions per 1000 prescriptions written. Using a Poisson model for probability, with $\lambda = 0.06$, find the probability of

- Exactly one gastrointestinal reaction in 1000 prescriptions. [Ans: 0.05651]
- No gastrointestinal reactions in 1000 prescriptions. [Ans: 0.94176]
- Exactly two gastrointestinal reactions in 1000 prescriptions. [Ans: 0.00170]
- At least one gastrointestinal reaction in 2000 prescriptions. [Ans: 0.11308 (when $\lambda = 0.12$)]

Question 6. A private clinic monitors the number of emergency patients it receives daily. Over time, it has been observed that the clinic averages 10 emergency patients per day. The arrival of these patients is random and follows a pattern that can be modeled using the Poisson distribution, which is commonly used to predict the probability of a given number of events happening within a fixed interval of time when the events occur independently.

- What is the probability that the clinic will receive exactly 5 emergency patients on a specific day? [Ans: 0.03783]
- What is the probability that the clinic will receive at least 3 emergency patients on a specific day? [Ans: 0.997]
- What is the probability that the clinic will receive between 5 and 10 emergency patients (inclusive) on a specific day? [Ans: 0.554]

Question 7. A group of BRAC University students created a ride-share app where customers and riders can negotiate fares. Their helpline center received an average of 6 calls per hour related to software bugs.

- Write down the probability mass function of the distribution. $\left[P(X) = \frac{e^{-6}6^x}{x!}\right]$
- What is the probability that they will receive exactly 8 calls per hour? [Ans: 0.10]
- What is the probability that they will receive at most 4 calls per hour? [Ans: 0.2851]
- What is the probability that they will receive more than 3 calls per hour? [Ans: 0.8488]

Question 8. On average, a factory experiences 2 machine failures per day.

- What is the probability that the factory will have no machine failures on a given day? [Ans: 0.1353]
- Find the probability of having at least three machine failures on a given day. [Ans: 0.32332]
- Find the probability of having at most two machine failures on a given day. [Ans: 0.67668]
- Determine the probability of having exactly 5 machine failures on a given day. [Ans: 0.03609]

Question 9.

According to the National Office of Vital Statistics of the Vancouver Health and Human Services, the average number of accidental drownings per year in the state is 3.0 per 10,000 populations. Find the probability that in a city of population 20,000, there will be-

- 2 accidental drownings per year. [Ans: 0.0446]
- At least 3 accidental drowning per year. [Ans: 0.938]
- Between 4 and 8 accidental drowning per year. [Ans: 0.6964]

Solution:

Let, X be the random variable representing number of accidental drownings per year

Here, Average number of accidental drownings per year, $\lambda = 200000 \times \frac{3}{100000} = 6$

So, $X \sim \text{Poisson}(\lambda = 6)$

- Try yourself.
- Try yourself.
- Try yourself.

Question 10. In a vintage vinyl record factory in Bangladesh, the quality control team has observed that the number of defects per record follows a Poisson distribution with a mean of 0.5 defects per record. The factory produces limited edition records that are highly valued by collectors. The management aims to ensure that each record meets high-quality standards before being sold.

- Determine the probability that a randomly selected record will have exactly two defects. [Ans: 0.0758]
- Given that the factory wants to ensure all records sold to customers are defect-free, calculate the percentage of records that will be rejected for sale because they have defects. **(Hint: Use $e^{-0.5} \approx 0.6065$ for your calculations.)** [Ans: 39.35%]
- Suppose the factory wants to set a new quality standard where only records with at most one defect are allowed for sale. Calculate the probability that a randomly selected record will meet this new standard. [Ans: 0.9098]
- Calculate the variance and standard deviation of the number of defects per record in the factory's production.

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Geometric Distribution

Question 1. The probability of a successful optical alignment in the assembly of an optical data storage product is 0.8. Assume the trials are independent.

- What is the probability that the first successful alignment requires exactly four trials?
- What is the probability that the first successful alignment requires at most four trials?
- What is the probability that the first successful alignment requires at least four trials?

Solution: Consider the trials are independent and the probability of success (successful optical alignment) is $p = 0.8$. We define X as the number of trials until the first success, so X follows a geometric distribution with $p = 0.8$. i.e., $X \sim \text{Geom}(0.8)$ and the probability mass function (pmf) of X is given by: $P(X = x) = 0.8 \times (1 - 0.8)^{x-1}$ for $x = 1, 2, 3, \dots$

- The probability that the first successful alignment requires exactly four trials:

$$\text{For } x = 4: P(X = 4) = 0.8 \times (1 - 0.8)^{4-1} = 0.0064$$

So, the probability that the first successful alignment requires exactly four trials is 0.0064.

- The probability that the first successful alignment requires at most four trials:

This is the cumulative probability of the first success occurring within the first four trials:

$$P(X \leq 4) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

Calculating each:

$$P(X = 1) = 0.8 \times (0.2)^0 = 0.8$$

$$P(X = 2) = 0.8 \times (0.2)^1 = 0.16$$

$$P(X = 3) = 0.8 \times (0.2)^2 = 0.032$$

$$P(X = 4) = 0.8 \times (0.2)^3 = 0.0064$$

Adding these together:

$$P(X \leq 4) = 0.8 + 0.16 + 0.032 + 0.0064 = 0.9984$$

So, the probability that the first successful alignment requires at most four trials is 0.9984.

- The probability that the first successful alignment requires at least four trials. This is the complementary probability of the first success happening after three trials:

$$P(X \geq 4) = 1 - P(X \leq 3)$$

$$\text{From part (b): } P(X \leq 3) = 0.8 + 0.16 + 0.032 = 0.992$$

$$\text{Therefore: } P(X \geq 4) = 1 - 0.992 = 0.008$$

So, the probability that the first successful alignment requires at least four trials is 0.008.

Question 2. In the programming club at BRAC University, members occasionally encounter bugs while working on projects. It is known from previous experience that the probability of encountering a bug during any coding session is 3%. To evaluate their debugging process, the club decides to monitor the occurrence of bugs during their coding sessions.

- What is the probability that the first bug is encountered during the 6th coding session?
- What is the probability that the first bug is found within the first 3 coding sessions?
- On average, how many sessions would you expect to conduct before encountering the first bug?
- Calculate the variance and standard deviation of the number of sessions until the first bug is found.
- Calculate the coefficient of variation (CV) for the number of sessions until the first bug is found.

Solution: Let X be the number of coding sessions until the first bug is encountered. Since the probability of encountering a bug during any coding session is $p = 0.03$, X follows a geometric distribution with parameter p .

- The probability that the first bug is encountered during the 6th session is given by:

$$P(X = 6) = (1 - p)^{6-1} \times p = (1 - 0.03)^5 \times 0.03 = 0.0251$$

So, the probability is 0.0251.

- The probability that the first bug is found within the first 3 sessions is:

$$P(X \leq 3) = P(X = 1) + P(X = 2) + P(X = 3)$$

$$P(X = 1) = 0.03$$

$$P(X = 2) = 0.97 \times 0.03 = 0.0291$$

$$P(X = 3) = (0.97)^2 \times 0.03 \approx 0.941 \times 0.03 \approx 0.0274$$

$$P(X \leq 3) = 0.03 + 0.0291 + 0.0282 = 0.0873$$

So, the probability is 0.0873.

- The expected value (mean) of X for a geometric distribution is:

$$E(X) = \frac{1}{p} = \frac{1}{0.03} \approx 33.33$$

So, on average, you would expect to conduct 33.33 sessions before encountering the first bug.

- The variance of X for a geometric distribution is:

$$Var(X) = \frac{1 - p}{p^2} = \frac{1 - 0.03}{(0.03)^2} = \frac{0.97}{0.0009} \approx 1077.78$$

The standard deviation is the square root of the variance:

$$SD(X) = \sqrt{1077.78} \approx 32.82$$

- e. The coefficient of variation (CV) is given by:

$$CV = \frac{SD(X)}{E(X)} = \frac{32.82}{33.33} \approx 0.9847$$

So, the coefficient of variation is approximately 0.9847.

Question 3. Max decides to implement a quality control process to minimize the number of defective bulbs reaching customers. He instructs his team to inspect every batch of 75 bulbs and remove any defective ones before packaging. Unfortunately, he found that, on average, there were 3 defective bulbs in every batch.

- What is the probability that Max will find the first faulty lightbulb on the 6th one that he tested? [Ans: 0.0326]
- What is the probability that it takes at least six trials until he finds the first defective lightbulb?
- Find the number of lightbulbs we would expect Max to inspect until he finds his defective, as well as the standard deviation? [Mean = 25, SD = 24.5]

Question 4. A die is thrown until 6 occurs for the first time. What is the probability that

- Exactly 3 tosses will be required. [Ans: 0.69]
- More than two tosses will be required.
- Find the average number of tosses required. [Ans: 6]
- Find the variance of this distribution.

Question 5. The probability that a player qualifies in a national chess championship is 0.7.

- Find the probability that a player chosen at random qualifies the championship on the third attempt. [Ans: 0.063]
- Find the probability that a player chosen at random qualifies the championship before the fourth attempt. [Ans: 0.973]
- Determine the mean and standard deviation of the above distribution. [Mean = 1.428, SD = 0.78]